SECTION A — SCHEMA ANALYSIS AND NORMALIZATION

Given Schema

```
PatientRecords (
  patient_id,
  patient_name,
  patient_phone,
  doctor_name,
  doctor_specialization,
  appointment_date,
  diagnosis,
  treatment,
  prescription_details,
  medicine_name,
  medicine_price,
  bill_amount,
  payment_status
)
```

A1.Issues

1. It is not in normalized form
2. Values are being repeated as prescription_details contain multiple medicine for a one patient
3. Patient_id and Patient_name is being repeated for every doctor visit
4. There is no Doctor_id as Doctor_name may have same values
5. The schema poorly structure as some values are dependent on other values

   We can add these entity int the schema:

   - Doctor_id
   - Bill_id
   - Appointment_id

- Patient_age

A2. Normalize the Schema

1NF: Patients (
patient_id PK,
patient_name,
patient_phone
)

Doctors (
doctor_id PK,
doctor_name,
doctor_specialization
)

Appointments (
appointment_id PK,
patient_id FK,
doctor_id FK,
appointment_date,
diagnosis,
treatment
)

Prescriptions (
prescription_id PK,
appointment_id FK,
prescription_details
)

Medicines (
medicine_id PK,
prescription_id FK,
medicine_name,
medicine_price
)

Bills (

```
    bill_id PK,
    appointment_id FK,
    bill_amount,
    payment_status
)
```

2nf:

```
   Patients (
    patient_id PK,
    patient_name
)

Patient_Phones (
    phone_id PK,
    patient_id FK,
    patient_phone
)

Doctors (
    doctor_id PK,
    doctor_name
)

Doctor_Specializations (
    specialization_id PK,
    doctor_id FK,
    specialization_name
)

Appointments (
    appointment_id PK,
    patient_id FK,
    doctor_id FK,
    appointment_date
)

Diagnoses (
    diagnosis_id PK,
    appointment_id FK,
```

```
    diagnosis_details
)

Treatments (
    treatment_id PK,
    appointment_id FK,
    treatment_details
)

Prescriptions (
    prescription_id PK,
    appointment_id FK
)

Prescription_Details (
    prescription_id FK,
    medicine_id FK,
    quantity,
    PRIMARY KEY (prescription_id, medicine_id)
)

Medicines (
    medicine_id PK,
    medicine_name,
    medicine_price
)
```

3NF:

```
       Patients (
    patient_id PK,
    patient_name
)

Patient_Phones (
    phone_id PK,
    patient_id FK,
    patient_phone
)
```

```
Doctors (
    doctor_id PK,
    doctor_name
)

Doctor_Specializations (
    specialization_id PK,
    doctor_id FK,
    specialization_name
)

Appointments (
    appointment_id PK,
    patient_id FK,
    doctor_id FK,
    appointment_date
)

Diagnoses (
    diagnosis_id PK,
    appointment_id FK,
    diagnosis_details
)

Treatments (
    treatment_id PK,
    appointment_id FK,
    treatment_details
)

Prescriptions (
    prescription_id PK,
    appointment_id FK
)

Prescription_Details (
    prescription_detail_id PK,
    prescription_id FK,
    medicine_id FK,
    quantity
```

```
)

Medicines (
    medicine_id PK,
    medicine_name,
    medicine_price
)

Bills (
    bill_id PK,
    appointment_id FK,
    bill_amount
)

Payments (
    payment_id PK,
    bill_id FK,
    payment_status
)
```

A3. Assumptions

- Each patient may have more than one phone number so it is stored in a different table phone_id is used to identify the unique number and patient_id is used as the the foreign key
- Each doctor is given a unique id as names can be repeated
- Doctors_Specilization is stored in a separate table as a doctor can have more than one specialization and Doctor_id used as the foreign key
- Each appointment is between one patient and doctor
- Each diagnosis is linked to the one appointment only
- Created a medicine_id which is unique for a medicine name and also linked with the price of the medicine

SECTION B — DATABASE DESIGN FROM REQUIREMENTS

B1. Requirement Clarification

1. Does one patient have multiple phone number?
2. Can a doctor have multiple specializations?
3. Will a patient book multiple appointements with different doctor in a day?
4. Should prescriptions should be based on previous appointments?
5. Can multiple diagnosis be noted down in the treatment?
6. Should the appointments use fixed slots?

B2. Design

```
 Patients (
  patient_id PK,
  patient_name,

)

Patient_Phones (
  phone_id PK,
  patient_id FK,
  patient_phone
)

Departments (
  department_id PK,
  department_name
)

Doctors (
  doctor_id PK,
  doctor_name,
  department_id FK,

)

Doctor_Specializations (
```

```
    specialization_id PK,
    doctor_id FK,
    specialization_name
)

Appointments (
    appointment_id PK,
    patient_id FK,
    appointment_date,
    appointment_time,
    appointment_status
)

Consultations (
    consultation_id PK,
    appointment_id FK,
    consultation_date,
    observations,
    follow_up_of FK NULL
)

Consultation_Doctors (
    consultation_doctor_id PK,
    consultation_id FK,
    doctor_id FK,

role,
    contribution_notes
)

Diagnoses (
    diagnosis_id PK,
    consultation_id FK,
    diagnosis_details,
    diagnosis_type
)

Treatments (
    treatment_id PK,
    consultation_id FK,
    treatment_details,
    treatment_cost
)
```

```
Medicines (
   medicine_id PK,
   medicine_name,
   medicine_price
)

Prescriptions (
   prescription_id PK,
   consultation_id FK,
   prescription_date
)

Prescription_Details (
   prescription_detail_id PK,
   prescription_id FK,
   medicine_id FK,
   dosage,

   duration,
   quantity
)

Bills (
   bill_id PK,
   consultation_id FK,
   consultation_charge,
   treatment_charge,
   medicine_charge,
   bill_date
)

Payments (
   payment_id PK,
   bill_id FK,
   payment_date,
   payment_amount,
   payment_status
)
```

B3. Normalization

The above design uses 3nf normalization and the design is in already normalized

Update During Development

The hospital has introduced an operational change.

A single consultation can involve multiple doctors, and each doctor must be associated with their contribution to the consultation.

Before the design was in such a manner that one consultation can have only one doctor.But now the requirement arises that one consultation can have multiple doctors.
Also each doctor’s role in that consultation should be store.
One doctor can be the part of many consultations.

```
Consultation_Doctors (
   consultation_doctor_id PK,
   consultation_id FK,
   doctor_id FK,
   contribution_role,
   contribution_notes
)
```

SECTION C — QUERIES AND TRANSACTIONS

The queries and transactions code is present in the sql file